CISC7204 Data Science and Data Visualization

2026/2027, Semester 1 | Class G - Team 03

TEAM WORK PLAN 02

Item	Details
Project title	The Power-Hungry World: Comparing Global Electricity Consumption Across Sectors
Submission date	10 November 2026
Project start	5 November 2026
Analysis scope	Global final electricity consumption, 1990-2022
Status date	10 November 2026; current working version

Team members

Full name	Student ID	Primary responsibility
UN CHI KIT	mc653081	Technical development, analysis and technical outputs
KELDEN JAMTSHO	mc664183	Data collection and preprocessing
GAN RUIYAN	mc664801	Presentation PPT preparation
HE YIFEI	mc664916	Progress documentation and document organization
SHENG TIAN	mc664961	Presentation PPT preparation
TANG JING	mc666256	Progress records, meeting minutes and document organization

The team has established a working data and visualisation workflow. This plan records implemented outputs, ongoing review and the remaining work; it does not represent the project as fully finalised.

Team 03 will participate in the instructor-assigned presentation session on either 18 or 25 November 2026. The team is preparing the required presentation materials accordingly.

1. Project scope and data source strategy

Research question: How has final electricity consumption changed across countries and regions between 1990 and 2022, and which countries and end-use sectors account for the largest shares of that consumption? The title, research questions, global scope and analytical period remain unchanged; only the dataset strategy and preparation procedures have changed.

Date	IEA correspondence and resulting strategy
5 Nov	Kelden requested permission for public academic outputs. IEA prohibited dataset downloads via GitHub and requested details of derived-data use.
6-7 Nov	Kelden clarified the intended public tables, charts, maps and dashboard code. IEA asked whether underlying values could be extracted or reverse-engineered.
9 Nov	IEA confirmed that the proposed public use was not permitted because underlying values could be inferred or recovered, and advised an alternative source if adaptation was infeasible.
As of 10 Nov	IEA licensing restrictions led to the UNdata-primary workflow. Official national supplements include AGEB for German gaps and a bank report for Pakistan 1990-1993; OWID and other suitable sources support validation and context. Integration and validation continue.

2. Project progress as of 10 November

Area	Current working outputs	Ongoing review or next work
Acquisition and preparation (Supplementary national sources)	UN cleaned data and TWh analysis tables implemented. AGEB German gap-filling completed; A State Bank of Pakistan report was reviewed for historical data covering 1990–1993. Historical corrections for 1990–1992 have been applied in the current data pipeline, while the 1993 source comparison is documented in the supplementary notebook. Supplementary sources: France: UNdata complete under code 251 (France-Monaco); SDES 2026 kept as a cross-check. Indonesia: ESDM HEESI transport 2000–2007 and 2016 extracted; matches UNdata exactly for 2008–2014. Saudi Arabia: GASTAT 2010–2023 sector data extracted.	Retain AGEB/bank-report provenance. Indonesia integration and estimate sensitivity remain ongoing.
Country and regional mapping	6,932 country-years; five regions plus World. Country codes and combined areas mapped; 214/218 polygon metadata matches.	Historical coverage varies; unmatched territories and map coverage need disclosure.
EDA and trends	Saved notebook gap/source comparisons; global and regional trend figures, including a 179-area fixed sample.	Further EDA, growth measures and coverage-aware interpretation.
Rankings and sector composition	2022 Top 10, endpoint country shares and four-sector global bars implemented.	Pakistan transport zeros are historical values per the technical lead; retain evidence and review sector comparability and regional results.
Geographical visualisation	1990/2022 maps and animated Plotly slider exist.	Map matching, captions, rendering and interaction checks.
Validation and dashboard	Validation scripts and stored flags; static Plotly website with charts and CSV downloads.	Cross-source flags, estimate sensitivity, technical testing and possible dashboard refinement.
Presentation preparation	Proposal and introductory video materials exist; group photo previously submitted as reported by the team.	Prepare/update proposal, photo, video and separate WIP PPT for the assigned session; WIP completion is not claimed.

3. Team structure and assigned responsibilities

The primary responsibilities below govern the current work plan. Supporting duties and future deliverables are current assignments and planned contributions. Major methodological decisions remain with the team.

Member	Primary responsibility	Supporting duties and next deliverables
KELDEN JAMTSHO mc664183	Data collection and preprocessing	Evaluate/collect sources; preprocess UNdata and national extracts; maintain AGEB/Pakistan provenance; assess missingness, sector definitions and quality; coordinate Indonesia integration review.
UN CHI KIT mc653081	Technical development, analysis and technical outputs	Implement analysis/calculations and analysis tables; maintain figures, maps and dashboard; integrate agreed data updates; conduct technical testing and sensitivity calculations.
HE YIFEI mc664916	Progress documentation and document organization	Structure and integrate reports; organise source/evidence records; coordinate presentation narrative and team communication; obtain instructor scheduling information.
TANG JING mc666256	Progress records, meeting minutes and document organization	Maintain minutes, progress/action records and shared documentation; track deliverables; coordinate document review and submission checks.
SHENG TIAN mc664961	Presentation PPT preparation	Develop slide visual design, chart layouts and visual consistency; check readability and support rehearsal for the assigned presentation.
GAN RUIYAN mc664801	Presentation PPT preparation	Structure PPT content and storyline; integrate technical charts with the technical lead; explain qualified analytical findings and limitations.

Evidence and contribution recording

Kelden is credited in the source-evaluation draft and supplementary report, and the email record documents his IEA correspondence. Meeting 01 credits Tang Jing as minute preparer; introductory materials name He Yifei as presenter. Current technical updates are provided by UN CHI KIT. Team outputs are not automatically attributed to individual authors.

Outstanding technical review

Indonesia supplementary values remain unintegrated in the current analysis tables. Estimate-exclusion sensitivity and cross-source validation remain ongoing; dashboard refinement may be needed. Pakistan transport zeros are genuine historical values according to UN CHI KIT, who attributes them to fossil-fuel and charcoal use in trains and cars during those years. Retain the supporting source evidence and ordinary validation checks.

4. Working schedule from 5 November

The following target dates guide work from 5 November to 3 December. Presentation materials will be prepared for either instructor-assigned session, with updates for the later session where applicable.

Period / target	Work and expected evidence	Lead / support
5-10 Nov Current progress	Source/licence research and working pipeline, tables, figures, maps and site. Record AGEB filling, Pakistan bank-report use and current progress.	Kelden / Un; He and Tang organise records
By 17 Nov Review target	Complete remaining source/preprocessing review; address Indonesia integration; retain AGEB and Pakistan source evidence and documented transport-zero explanation.	Kelden / Un
By 17 Nov Review target	Review cross-source flags, estimate sensitivity, calculations, figures/maps and dashboard tests; document remaining work for presentation.	Un / Kelden
17 Nov; update 24 Nov if needed	Prepare/update proposal, photo, introductory video and WIP PPT for the assigned session; integrate reviewed charts and limitations.	Gan/Sheng: PPT; He/Tang: package; Un/Kelden: review
17 Nov; repeat 24 Nov if needed	Rehearse, check readability and timing; prepare Public Forum upload for noon on the assigned 18 or 25 Nov session.	All; He coordinates; Tang checks
18 or 25 Nov	Participate in the instructor-assigned presentation session and record feedback.	All members
By 25 Nov	Refine analysis/visual outputs and dashboard as needed; incorporate available review/presentation feedback.	Un / Kelden; Gan/Sheng: PPT updates
26 Nov-3 Dec Proposal target	Finalise report, validation documentation, citations, figures and review; prepare main-project submission by 3 Dec.	He: integration; Tang: checks; all review

Presentation package

Team 03 will participate in the instructor-assigned presentation session on either 18 or 25 November 2026. The team is preparing the required presentation materials accordingly.

The package comprises the proposal, group photo, introductory video and separate WIP PPT, to be prepared or updated for the assigned session. Introductory-video slides are distinct from the WIP PPT, which remains a planned deliverable. Public Forum submission is targeted for noon on the assigned session date; no 11 November presentation deadline applies.

5. Planned regular team meetings

Subsequent regular meetings are online on Wednesdays, 16:00-16:30 (Asia/Shanghai). TANG JING is the minute taker. Wednesday evening classes run from 19:00 to 22:00, so no evening team meetings are scheduled.

Date	Purpose	Minute taker
11 Nov 2026	Review data/validation issues, current outputs and presentation allocation.	Tang Jing
18 Nov 2026	Review analytical updates and presentation readiness or feedback, according to assignment.	Tang Jing
25 Nov 2026	Review presentation readiness or feedback, report and dashboard progress.	Tang Jing
2 Dec 2026, if required	Final checks, remaining actions and main-submission readiness.	Tang Jing

6. Risks and quality assurance

Risk / issue	Mitigation or review activity	Responsible
Licensing restrictions	Use the UN-led strategy and permitted materials; retain citations/licence records; do not upload restricted IEA raw data.	Kelden; He/Tang organise records
Supplementary sources and gaps	Retain AGEB and Pakistan bank-report provenance, year mapping and source evidence; review Indonesia integration and disclose remaining gaps.	Kelden / Un
Comparability and units	Check consumption versus demand/generation, combined areas, sectors and TWh conversions; retain the technical lead's historical explanation and evidence for Pakistan transport zeros.	Kelden / Un
Estimates and validation uncertainty	Complete appropriate sensitivity and cross-source review; report flags and uncertainty without unsupported causal explanations.	Un / Kelden
Dashboard and figures	Test working static Plotly site, maps, slider, downloads, labels and links; refine as necessary. No completed Dash implementation is claimed.	Un; Gan/Sheng check presentation use
Presentation date and package	Await instructor assignment; keep package ready for either possible session; verify forum timing and final review.	He / Tang; Gan/Sheng prepare PPT

7. Review and submission checks

All six Team 03 members have reviewed and approved this Work Plan 02. Tang Jing maintains the action/submission record and He Yifei coordinates document integration. Approval covers the project plan and responsibilities; the technical review and development tasks identified above remain ongoing.

Evidence: Project_Audit_20261110 status and contribution reports; original proposal and Work Plan 01; Meeting 01 minutes; supplementary report/notebook; team03/DATA_PIPELINE.md; IEA email record, 5-9 November 2026. Technical outputs remain working versions.